

Harmanpreet Singh Sagar

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EDUCATION

McMaster University

Sept. 2022 - June 2027

B.Eng., Software Engineering (Co-op)

GPA: 3.83/4.0

Awards: Dean's Honour List (3x), Lawrence Tse Eng. Research Award (\$6,000), A.G. Reilly Scholarship (\$5,000), McMaster University Award of Excellence

Relevant Courses: Software Design, Data Structures and Algorithms, Databases, Operating Systems, Software Testing

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, Java, C, Embedded C, C#, Haskell, SQL, HTML, CSS, Bash

Frameworks and Libraries: React, Next.js, FastAPI, Node.js, Tailwind CSS, FreeRTOS, libcs, Pydantic, Restinio, Firebase, JUnit, Pytest

Developer Tools: Git, Linux, Docker, Jenkins, GitHub Actions, PostgreSQL, Supabase, CMake, Maven, Perforce, MATLAB, Simulink, Vitis, REST APIs, CAN, MQTT

EXPERIENCE

Synopsys Inc.

May 2024 - Aug. 2025

System Architecture and Modelling Engineering Intern

Mississauga, ON

- Cut broken builds reaching the **Perforce** depot by **90%** with a two-tier pre-commit and nightly regression system.
- Reduced the pre-commit gate runtime by **55% (40 mins to 18 mins)** by parallelizing **Jenkins CI/CD** stages.
- Decreased **SERDES** debug time by **50%** by building an HTTP-based trace-observer tool with **Restinio** in **C++**.
- Verified high-speed SERDES designs in **C++**, achieving **100% code and functional coverage**.

McMaster Interdisciplinary Satellite Team

Sept. 2023 - Present

Firmware Specialist, PRESET CubeSat

Hamilton, ON

- Implemented a **CubeSat Space Protocol (CSP)** server in **Embedded C** on a **Xilinx Zynq ZedBoard** under **FreeRTOS**, running over **2 transports** (CAN bus and ground umbilical).
- Built **5 CSP service handlers** (firmware update, **AXI** register access, FSM, CLI, science data) and ported the stack across **2 board revisions**.
- Expanded payload-CDH control to **14 commands** with a **UART-backed** command server and **CLI**.

Develop for Good

May 2026 - Aug. 2026

Web Developer

San Francisco, CA (Remote)

- Evaluated **4 build approaches** and recommended **Webflow** for CMS-driven maintainability by a non-technical client.
- Built and redesigned **3 of 8 site pages** on a **3-engineer team**, implementing **Figma** designs responsively.

McMaster University, McSCert

Sept. 2026

Incoming Research Assistant, jPipe (Part-time)

Hamilton, ON

- Selected to contribute to a **Java** and **ANTLR4** DSL compiler for safety-critical software certification.

McMaster University, Department of Computing and Software

May 2023 - Aug. 2023

Research Assistant, *Drasil*

Hamilton, ON

- Strengthened consistency checks across a **200k+ line** codebase, cutting functional duplication by **25%**.
- Used **Haskell** to produce consistent object-oriented code generation across **5+ case studies**.

PROJECTS

F1 Battle Detector | [Code](#)

Python, FastAPI, Next.js, TypeScript, Docker

- Built a **real-time** Formula 1 battle detector in **Python** and **FastAPI**, fusing **3 live OpenF1 streams** across **20 drivers** on a **1.5s** loop.
- Ranked battles across **3 weighted signals** (gap, closing rate, pace delta) with **hysteresis** filtering to suppress false positives.
- Built a **Next.js** dashboard and cut its container image **82% (1.19GB to 210MB)** with a multi-stage **Docker** build, backed by **62 tests** in **GitHub Actions** CI.

RBC-BizConnect | [Website](#) | [Code](#)

Python, JavaScript, SQLite3, Firebase

- Won **1st place** in the RBC Challenge at **DeltaHacks IX** with a scalable database and **Firebase** authentication.
- Cut investor-business matching from **quadratic to linear time** with a single-pass algorithm, letting the platform scale to large datasets.